
CS 421 --- Type Semantics Activity (Polytype Version)

Name	Netid

Please write your name/netid legibly in dark ink. Hand in one copy per team. Do not staple or mangle the corners.

Objectives

When you are done with this activity, you will have

- Described the $\geq$ relation on types.
- Described the relation between the GEN and INST rules with $\forall$.
- Used a proof tree to show the difference between monomorphic application and polymorphic **let**.

Note that the list of rules can be found at the end of the problems.

The $\geq$ Operation for Types

Time estimate: 5 -- 10 minutes

Instructions Consider the following table of $\geq$ relations. Some are marked ``valid" and some are marked ``invalid."

$\forall\alpha.\alpha \rightarrow \alpha$	$\geq$	$\text{Int} \rightarrow \text{Int}$	valid
$\forall\alpha.\forall\beta.\alpha \rightarrow \beta$	$\geq$	$\forall\beta.\text{Int} \rightarrow \beta$	valid
$\forall\alpha.\alpha \rightarrow \alpha$	$\geq$	$\forall\beta.\beta \rightarrow \beta$	valid
$\forall\alpha.\alpha \rightarrow \alpha$	$\geq$	$\text{Int} \rightarrow \text{String}$	invalid
$\forall\alpha.\alpha \rightarrow \alpha$	$\geq$	$\forall\alpha.\forall\beta.\beta \rightarrow \alpha$	invalid

Problem 1) Review the relations above and try to come up with a rule that describes when the $\geq$ relation is valid.

Problem 2) Is the relation $\forall\alpha.\forall\beta.\alpha \rightarrow \beta \geq \text{Int} \rightarrow \text{Int}$ a valid one?

1 GEN and INST

Time estimate 10--15 minutes.

Problem 3) Consider the following two proof trees.

$$\frac{\frac{\Gamma \vdash id : \mathbf{int} \rightarrow \mathbf{int}}{\Gamma \vdash id : \mathbf{int}} \text{VAR}}{\Gamma \vdash id x : \mathbf{int}} \text{APP} \quad \frac{\Gamma \vdash x : \mathbf{int}}{\Gamma \vdash id x : \mathbf{int}} \text{VAR}$$

$$\frac{\frac{\Gamma \vdash id : \forall \alpha. \alpha \rightarrow \alpha}{\Gamma \vdash id : \mathbf{int}} \text{VAR}}{\Gamma \vdash id x : \mathbf{int}} \text{APP} \quad \frac{\Gamma \vdash x : \mathbf{int}}{\Gamma \vdash id x : \mathbf{int}} \text{VAR}$$

One of these trees is not correct. Which one? Show how to fix it using the INST rule.

Problem 4) Consider this proof tree:

$$\frac{\frac{\frac{\{x : \alpha\} \vdash x : \alpha}{\{x : \alpha\} \vdash \lambda x. x : \alpha \rightarrow \alpha} \text{VAR}}{\{x : \alpha\} \vdash \lambda x. x : \forall \alpha. \alpha \rightarrow \alpha} \text{ABS}}{\Gamma \equiv \{\} \vdash \mathbf{let} \ f = \lambda x. x \ \mathbf{in} \ f \ 10 : \mathbf{int}} \text{GEN} \quad \frac{\frac{\frac{\{f : \forall \alpha. \alpha \rightarrow \alpha\} \vdash f : \forall \alpha. \alpha \rightarrow \alpha}{\{f : \forall \alpha. \alpha \rightarrow \alpha\} \vdash f : \mathbf{int} \rightarrow \mathbf{int}} \text{VAR}}{\{f : \forall \alpha. \alpha \rightarrow \alpha\} \vdash f \ 10 : \mathbf{int}} \text{INST}}{\Gamma \equiv \{\} \vdash \mathbf{let} \ f = \lambda x. x \ \mathbf{in} \ f \ 10 : \mathbf{int}} \text{LET} \quad \frac{\frac{\{f : \forall \alpha. \alpha \rightarrow \alpha\} \vdash 10 : \mathbf{int}}{\{f : \forall \alpha. \alpha \rightarrow \alpha\} \vdash f \ 10 : \mathbf{int}} \text{CONST}}{\Gamma \equiv \{\} \vdash \mathbf{let} \ f = \lambda x. x \ \mathbf{in} \ f \ 10 : \mathbf{int}} \text{APP}$$

Consider the GEN step. How do you know when you need to use it?

Proofs

Time estimate: 25 minutes.

Create proofs for the following judgements according to the given rules.

Problem 5) $\{\text{id}: \forall \alpha. \alpha \rightarrow \alpha, y: \text{Int}\} \vdash \text{id } y : \text{Int}$

Problem 6) $\{y: \text{Int}, z: \text{String}\} \vdash (\lambda f. (f \ y, f \ z)) (\lambda x. x) : (\text{Int}, \text{String})$

Problem 7) $\{x: \text{Int}, y: \text{String}\} \vdash \text{let } f = \lambda x. x \text{ in } (f \ x, f \ y) : (\text{Int}, \text{String})$

The Rules

Constants

$$\frac{}{\Gamma \vdash n : \mathbf{int}} \text{CONST, when } n \text{ is an integer.}$$

Similarly for True and False.

Variables

$$\frac{}{\Gamma \vdash x : \sigma} \text{VAR, when } x : \sigma \in \Gamma$$

Binary Arithmetic

$$\frac{\Gamma \vdash e_1 : \mathbf{int} \quad \Gamma \vdash e_2 : \mathbf{int}}{\Gamma \vdash e_1 \oplus e_2 : \mathbf{int}} \text{BINOP}$$

Integer Relations

$$\frac{\Gamma \vdash e_1 : \mathbf{int} \quad \Gamma \vdash e_2 : \mathbf{int}}{\Gamma \vdash e_1 \sim e_2 : \mathbf{bool}} \text{RELOP}$$

If

$$\frac{\Gamma \vdash e_1 : \mathbf{bool} \quad \Gamma \vdash e_2 : \tau \quad \Gamma \vdash e_3 : \tau}{\Gamma \vdash \mathbf{if } e_1 \mathbf{ then } e_2 \mathbf{ else } e_3 : \tau} \text{IF}$$

Application

$$\frac{\Gamma \vdash e_1 : \tau \rightarrow \tau' \quad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \tau'} \text{APP}$$

Abstraction

$$\frac{\Gamma \cup \{x : \tau\} \vdash e : \tau'}{\Gamma \vdash \lambda x. e : \tau \rightarrow \tau'} \text{ABS}$$

Let

$$\frac{\Gamma \vdash e_1 : \sigma \quad \Gamma \cup [x : \sigma] \vdash e_2 : \tau}{\Gamma \vdash \mathbf{let } x = e_1 \mathbf{ in } e_2 : \tau} \text{LET}$$

Letrec

$$\frac{\Gamma \cup [x : \tau_1] \vdash e_1 : \sigma \quad \Gamma \cup [x : \sigma] \vdash e_2 : \tau_2}{\Gamma \vdash \mathbf{letrec } x = e_1 \mathbf{ in } e_2 : \tau} \text{LETREC, where } \sigma \geq \tau_1$$

Gen

$$\frac{\Gamma \vdash e : \sigma}{\Gamma \vdash e : \forall \alpha. \sigma} \text{GEN, where } \alpha \text{ is not free in } \Gamma$$

Inst

$$\frac{\Gamma \vdash e : \sigma'}{\Gamma \vdash e : \sigma} \text{INST, where } \sigma' \geq \sigma$$